



CRITIC2: A program for real-space analysis of quantum chemical interactions in solids[☆]



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ABSTRACT

We present CRITIC2, a program for the analysis of quantum-mechanical atomic and molecular interactions in periodic solids. This code, a greatly improved version of the previous CRITIC program (Otero-de-la Roza et al., 2009), can: (i) find critical points of the electron density and related scalar fields such as the electron localization function (ELF), Laplacian, ... (ii) integrate atomic properties in the framework of Bader's Atoms-in-Molecules theory (QTAIM), (iii) visualize non-covalent interactions in crystals using the non-covalent interactions (NCI) index, (iv) generate relevant graphical representations including lines, planes, gradient paths, contour plots, atomic basins, ... and (v) perform transformations between file formats describing scalar fields and crystal structures. CRITIC2 can interface with the output produced by a variety of electronic structure programs including WIEN2k, elk, PI, abinit, Quantum ESPRESSO, VASP, Gaussian, and, in general, any other code capable of writing the scalar field under study to a three-dimensional grid. CRITIC2 is parallelized, completely documented (including illustrative test cases) and publicly available under the GNU General Public License.

Program summary

Program title: CRITIC2

Catalogue identifier: AECB_v2_0

Program summary URL: http://cpc.cs.qub.ac.uk/summaries/AECB_v2_0.html

Program obtainable from: CPC Program Library, Queen's University, Belfast, N. Ireland

Licensing provisions: yes

No. of lines in distributed program, including test data, etc.: 11686949

No. of bytes in distributed program, including test data, etc.: 337020731

Distribution format: tar.gz

Programming language: Fortran 77 and 90.

Computer: Workstations.

Operating system: Unix, GNU/Linux.

Has the code been vectorized or parallelized?: Shared-memory parallelization can be used for most tasks.

Classification: 7.3.

Catalogue identifier of previous version: AECB_v1_0

Journal reference of previous version: Comput. Phys. Comm. 180 (2009) 157

Nature of problem:

Analysis of quantum-chemical interactions in periodic solids by means of atoms-in-molecules and related formalisms.

[☆] This paper and its associated computer program are available via the Computer Physics Communication homepage on ScienceDirect (<http://www.sciencedirect.com/science/journal/00104655>).

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Solution method:

Critical point search using Newton's algorithm, atomic basin integration using bisection, qtree and grid-based algorithms, diverse graphical representations and computation of the non-covalent interactions index on a three-dimensional grid.

Additional comments:

!!!! The distribution file for this program is over 330 Mbytes and therefore is not delivered directly when download or Email is requested. Instead a html file giving details of how the program can be obtained is sent. !!!!

Running time:

Variable, depending on the crystal and the source of the underlying scalar field.

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1. Introduction

Understanding the intermolecular interactions that rationalize the geometry and energetics of a crystal is not a simple task. At present, the toolbox to address this problem includes band structure calculations, (projected) density of states representations, and plots of the electron density or some subset of the states in the valence. Techniques based on the analysis of scalar fields (such as the density) in real-space are commonplace in gas-phase molecular studies (see [1] and references therein) and are increasingly popular in the solid state [2,3], particularly when applied to the calculation of atomic charges [4,5].

The Quantum Theory of Atoms in Molecules (QTAIM) [6,7,1] has been widely used to study chemical bonding in molecules. QTAIM is based on a partition of real space into atomic regions, called basins. The delimiting surfaces of these basins (the interatomic surfaces, IAS) are determined solely by the electron density (ρ) via the zero-flux condition:

$$\nabla\rho(\mathbf{r}) \cdot \mathbf{n}(\mathbf{r}) = 0 \quad (1)$$

where $\mathbf{r}$ is any point on the surface and $\mathbf{n}(\mathbf{r})$ is the vector normal to it. The zero-flux condition ensures that the atomic expectation value of momentum-based operators (e.g. the kinetic energy) obtained by integration over the basin are uniquely defined. In addition to the kinetic energy, the atomic contributions to any extensive property can be found by basin integration, including volumes, charges, magnetic moments and susceptibilities, polarizabilities, and energies [3,8–17]. The atomic properties most widely used are the atomic charges, which have been extensively used in the study of charge transfer in surface adsorption problems (for instance, [4,5]), undoubtedly thanks to the excellent performance of grid-based integration methods [18–21].

In QTAIM, special meaning is attributed to the critical points (CP) of the electron density, that is, the points $\mathbf{r}$ where $\nabla\rho(\mathbf{r}) = \mathbf{0}$. By Eq. (1), all CPs, except maxima, are located on an interatomic surface. The maxima of the electron density are almost always at the nuclei (except for non-nuclear maxima [22,23]) and are called nuclear critical points (NCP). First-order saddle points (bond critical points, BCP) correspond to the definition of chemical bonds in QTAIM (which are usually coincident with traditional chemical bonds). Second-order saddle points and minima are called ring CPs (RCP) and cage CPs (CCP) respectively. The topology (the complete set of CPs) of the electron density and, particularly, the properties (density, Laplacian, ...) at the critical points have been correlated to chemical concepts such as bond strength or covalent character [6,7], and are widely used in the crystallography community [24–27].

The techniques applied to the electron density in QTAIM (localization of CPs and integration of basins) can be extended to any chemically-meaningful scalar field [28]. For instance, the electron localization function [29,30] (ELF) and the Laplacian of

the electron density [31–33] contain information about electron localization in real-space. Furthermore, the molecular electrostatic potential [34] (MEP) can be used to determine the nucleophilic and electrophilic regions of a molecule. The study of scalar fields different from the electron density using critical points and basin integrations is called Quantum Chemical Topology (QCT), a term coined by Popelier [35–37], and is a potentially interesting way to extract chemical information from these fields.

The calculation of the topology and the atomic observables involves complex algorithms and success is not guaranteed, particularly in large systems. To circumvent this problem for intermolecular interactions, the non-covalent index (NCI) plotting technique has been recently proposed by Johnson and collaborators [38–40]. In NCI, a grid of the reduced density gradient (RDG):

$$s = \frac{1}{2(3\pi^2)^{1/3}} \frac{|\nabla\rho|}{\rho^{4/3}} \quad (2)$$

encompassing the intermolecular region is calculated. The isosurfaces of low s , with the electron density color-mapped on them, have been shown to be a reliable indicator of the appearance of non-covalent interactions between molecules [38,39] and in crystals [40]. The domains enclosed by the isosurfaces of low- s map molecular contacts and recover the topology of the electron density in the limit of $s = 0$. The NCI domains have been linked to Pauli repulsion regions and hence to intermolecular attraction [40], an important feature to look for when examining close contacts in crystal packing [41].

A number of programs implementing several of the real-space techniques described above have been made available and described in the literature [42–51,18–20,39,52], but to our knowledge a program that can seamlessly handle the diverse crystal (CIF files, inputs and outputs from different codes, ...) and scalar field descriptions (three-dimensional grids, augmented plane-waves, local orbitals, ...) in periodic solids and apply common real-space techniques to them does not exist. The CRITIC2 code is a new version of the old CRITIC program that has been extensively redesigned to solve this problem.

CRITIC2 is the new an improved version of CRITIC [53,54]. The new code accepts the crystal structure and one or more scalar fields in formats native to a number of popular electronic structure programs (WIEN2k [55,56], elk [57], PI [58,59], abinit [60,61], Quantum ESPRESSO [62], VASP [63,64], Gaussian [65], ...), then provides the tools to perform QTAIM, QCT and NCI analysis on those fields. The list of capabilities includes: localization of critical points, integration of attractor basins (including but not limited to Bader charges), graphical representations, and a simple interface to the NCI plotting technique. CRITIC2 can also be used to interconvert between different crystal descriptions and to perform arithmetic operations on scalar fields. Regarding the integration of atomic properties (including charges), CRITIC2 implements the Yu and Trinkle (YT) algorithm [21], along with bisection and QTREE [66].

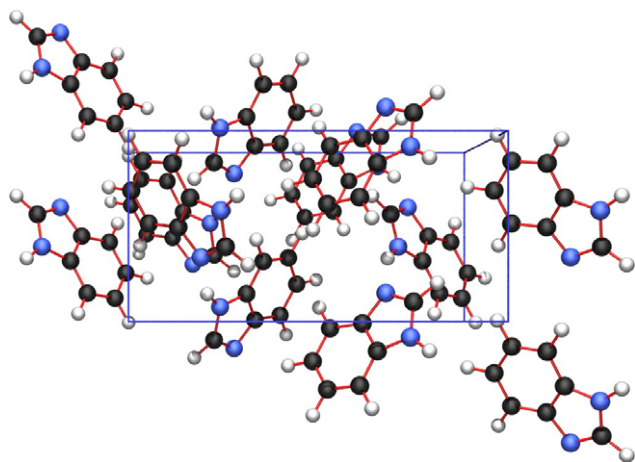


Fig. 1. The crystal structure of benzimidazole [68] at room temperature and pressure is shown. The space group is $Pna2_1$ (orthorhombic) with 4 molecules per cell. The asymmetric unit contains a single molecule.

YT and similar methods [18–20] are extremely efficient when the scalar field is given as a 3D grid. The new code is parallelized for shared-memory architectures, fully documented and released under an open license.

2. The crystal structure

The input for CRITIC2 is a single file, typically with extension `incritic`. This file contains a collection of keywords describing the crystal geometry, the source of the electron density, or any other scalar field, and the tasks to perform. The input is executed by doing:

```
critic2 file.incritic [file.outcritic]
```

with the output optionally written to `file.outcritic`.

The first few lines in the input usually define the structure of the crystal under study. CRITIC2 accepts a number of formats for the crystal geometry, including WIEN2k's `struct` format, crystallography information files (CIF), VASP's POSCAR-style files, Quantum ESPRESSO outputs, abinit densities (DEN), elk's `GEOMETRY.OUT`, and gaussian cube files. In addition, the geometry can also be specified by setting the cell parameters and either giving the space group and the list of atoms in the asymmetric unit, or by writing the list of all atoms in the cell. Symmetry operations are an important part of CRITIC2 because they are employed in various parts of the code to reduce the computational cost. Whenever the symmetry operations are not available in the input (for instance, a POSCAR), the internal symmetry finder, based on the original implementation by te Velde [67], is used.

We illustrate the input with a simple example: the benzimidazole crystal (Fig. 1). The crystal structure is passed to CRITIC2 using the `CRYSTAL` keyword. The source of the crystal geometry can be, for instance, a CIF file:

```
crystal BZDMAZ01.cif
```

or the files belonging to some electronic structure calculation:

```
crystal BZDMAZ01.struct # wien2k
crystal POSCAR H C N    # vasp
crystal BZDMAZ01.scf.out # espresso
crystal BZDMAZ01_o_DEN   # abinit
crystal GEOMETRY.OUT     # elk
crystal BZDMAZ01.cube    # others (gaussian cube)
```

The entry for VASP requires the specification of the atomic species (in the order in which they appear in the POTCAR) when the POSCAR does not contain this information.

In addition, we can pass the atomic positions and cell parameters manually by using the `CRYSTAL..ENDCRYSTAL` environment:

```
crystal
spg P n a 21
cell 25.52453 12.82935 13.11470 90 90 90
neq -0.257800 -0.395900 0.428800 C
... 13 atoms more ...
neq -0.282047 -0.651524 0.600529 H
endcrystal
```

The SPG keyword is followed by the space group label, which is parsed to generate the symmetry operations. The cell lengths (bohr) and angles (degrees) are read in CELL. The list of atoms in the asymmetric unit cell follows, entered using an instance of the NEQ keyword for each. The remaining atoms in the cell are automatically generated using the symmetry operations in the space group.

If the space group information is not available, one can resort to passing the complete list of atoms in the cell to `CRITIC2`. In that case, the internal symmetry finder will detect the symmetry operations and the lattice centering:

```
crystal
cell 13.507 6.789 6.940 90 90 90 ang
neq -0.257800 -0.395900 0.428800 C
... 58 atoms more ...
neq -0.282047 -0.651524 0.600529 H
endcrystal
```

In this case, CRITIC2 correctly detects that there are four non-trivial symmetry operations in the space group: 2, a , n , and the identity.

Finally, CRITIC2 can be used to interconvert between crystal structure formats using the `WRITE` keyword. For instance, a Quantum ESPRESSO input can be generated from a CIF file by running:

```
crystal BZDMAZ01.cif
write BZDMAZ01.scf.in
```

Input files for abinit, elk, VASP, and tessell [69] can also be created. A subset of the atoms in the crystal can be written to a molecular xyz file using:

```
write BZDMAZ01.xyz molmotif
```

The MOLMOTIF keyword instructs CRITIC2 to complete the molecules in the unit cell using atoms from the neighboring cells and a distance criterion based on the covalent radii. This is useful in two contexts: to generate a plot like the one shown in Fig. 1, and to define molecular fragments for NCIPLOT (Section 6).

3. Sources of the electron density and related scalar fields

CRITIC2 can read scalar fields from a variety of input sources, including WIEN2k's `clmsum`-style files, elk's `STATE.OUT`, PI's ion files and fields defined on a grid generated by VASP (CHGCAR), abinit (DEN), xcrysden [70] (xsf), and Quantum ESPRESSO/Gaussian (cube). In addition, most solid-state electronic structure codes can generate densities on a grid in one of those formats, which is useful in the case when CRITIC2 cannot interface their native formats directly. The internal representation of the field depends on its source: whenever possible, CRITIC2 uses the analytical representation of the field.

The `LOAD` keyword is used to read a field from an external file. The loaded fields are numbered in sequence as they appear in the

input, starting with 1. The promolecular density is available in slot 0 as soon as the crystal structure is specified. The syntax of LOAD is similar to CRYSTAL. For instance:

```
load file.cube
```

loads a gaussian cube file into the next available slot. In the following sections, we present the different sources and properties of the loaded fields available in CRITIC2. For clarity, we only consider the electron density, but the discussion applies to any other scalar field.

3.1. FPLAPW: WIEN2k and elk

In the full-potential linearized augmented plane-wave (FPLAPW) method [71,72], real space is divided into two regions: spheres around the nuclei (the muffin-tins, S_A for atom A) and the interstitial (I). The basis functions are expressed differently in each region: a spherical harmonics expansion inside the muffin-tins and a plane-wave expansion in the interstitial. This cuts down the number of basis functions required to represent the all-electron density and circumvents the use of pseudopotentials, at an increased computational cost and complexity. The density is written as:

$$\rho(\mathbf{r}) = \begin{cases} \sum_{LM}^{L_{\max}} \rho_{LM}(r') Y_L^M(\hat{r}'), & \mathbf{r} \in S_A, \\ \sum_{\mathbf{K}}^{K_{\max}} \rho_{\mathbf{K}} e^{i\mathbf{K} \cdot \mathbf{r}}, & \mathbf{r} \in I \end{cases} \quad (3)$$

where r' is the distance and $\hat{r}'$ are the angular coordinates referred to atom A , Y_L^M is a spherical harmonic function, $\mathbf{K}$ are reciprocal lattice vectors and ρ_{LM} and $\rho_{\mathbf{K}}$ are the density coefficients. The summations are truncated at the angular momentum $L_{\max}$ and the reciprocal vector norm $K_{\max}$. CRITIC2 reads the density coefficients from a file written by the FPLAPW code, together with all the ancillary information about the basis set (muffin-tin radii, $L_{\max}$, ...).

Two popular programs using the FPLAPW method are interfaced by CRITIC2: WIEN2k [55,56] and elk [57]. In WIEN2k the coefficients are written to files with different extensions depending on the field. For the density, this file has the extension `clmsum`, and is loaded in CRITIC2 with:

```
LOAD file.clmsum file.struct
```

In elk, the coefficients are written to a `STATE.OUT` file:

```
LOAD STATE.OUT GEOMETRY.OUT
```

In both cases, a pointer to some other file (`file.struct` or `GEOMETRY.OUT`) is necessary in order to locate the auxiliary structural information, such as the muffin-tin radii.

The internal representation of the FPLAPW density and its derivatives up to the second order is analytical (i.e. no numerical differentiation is necessary). This makes the localization of CPs and the integration of atomic basins easier. The drawback is that the spherical harmonics expansion in Eq. (3) is truncated at some $L_{\max}$ value, creating a discontinuity on the muffin-tin surface. The implications are: (i) the integral of the Laplacian in an atomic basin is not zero, and (ii) spurious CPs appear whenever the sign of the radial component of the gradient changes sign across the muffin-tin surface [33]. The latter interferes with the CP localization and gradient path tracing algorithms. In the particular case of the electron density, avoiding this problem is usually straightforward by using higher $L_{\max}$ and $K_{\max}$ in Eq. (3), although this increases the computational cost. In steeper fields, like the ELF, the discontinuity might be difficult to avoid. A simple check to detect spurious CPs on the muffin-tin surface is always performed by CRITIC2 when a FPLAPW field is loaded.

3.2. Ab-initio perturbed ion

The ab initio perturbed ion (aiPI) method [73–75] is based on the theory of electronic separability by Huzinaga et al. [76–79]. In aiPI, the system is partitioned into groups, which are assumed to be weakly interacting. As a consequence, the domain of applicability of aiPI is restricted to ionic crystals and closed-shell interactions. In aiPI, the many-electron wavefunction is written as the antisymmetrized product of ionic wavefunctions. Every ionic wavefunction is solved self-consistently under the effect of the rest of the atoms in the crystal, which is modeled by an external potential. The basis set is atom-centered and composed of exponential functions. The density is written as:

$$\rho(\mathbf{r}) = \sum_A \rho_A(\mathbf{r}) \quad (4)$$

where each of the atomic densities (ρ_A) is written as:

$$\rho_A(r) = \sum_{nl} n_{nl}^A \sum_{pq} c_{nlp}^A c_{nlq}^A \chi_{pl}^A(r) \chi_{ql}^A(r). \quad (5)$$

In this equation, n_{nl}^A are the orbital occupations of ion A (obtained by self-consistently solving the atomic problem for A), c_{nlp}^A are the orbital coefficients and χ_{pl}^A are the exponential functions. The ionic densities are spherical.

The PI program [58,59] is an implementation of the aiPI method which describes each ionic wavefunction by an `ion` file. These are loaded in CRITIC2 using:

```
LOAD file1.ion at1 file2.ion at2 ...
```

where the `at1, ...`, labels refer to either the atomic symbol or to the integer identifier of the atom within the cell. The aiPI electron density and its derivatives are analytical.

3.3. Plane-waves/pseudopotentials

In the pseudopotentials/plane-waves (PSPW) approach, the density is written as a plane-wave expansion:

$$\rho(\mathbf{r}) = \sum_{\mathbf{K}}^{K_{\max}} \rho_{\mathbf{K}} e^{i\mathbf{K} \cdot \mathbf{r}} \quad (6)$$

where $\mathbf{K}$ runs over reciprocal lattice vectors. The Kohn–Sham (KS) problem is solved in reciprocal space, but the $\rho_{\mathbf{K}}$ coefficients can be Fourier-transformed into real space to yield a uniform grid that spans the periodic cell. Because plane waves are not efficient at representing the steep features of the density, a pseudopotential that smooths out the oscillations in the one-electron states is always necessary. The pseudopotential also accounts for a number of core electrons which are frozen during the calculation. As a consequence, Eq. (6) represents a pseudo-density near the nuclei and integrates only to the number of valence electrons.

The PSPW method is the most popular condensed-matter electronic structure method at present, thanks to its accuracy and low computational cost, so many implementations are available. CRITIC2 reads density grids in several different formats:

```
LOAD file.cube # gaussian cube (espresso)
LOAD file_DEN # abinit
LOAD CHGCAR # vasp
LOAD file.xsf # xcrysden
```

which are natively written by Quantum ESPRESSO (or Gaussian), abinit, vasp and xcrysden [70].

In order to calculate the density and its derivatives at an arbitrary point in the cell, an interpolation method is necessary. We do this by using a three-dimensional cubic spline, which has been

adapted from the abinit code [60,61]. The density in Eq. (6) can be noisy, particularly in the interstitial, which disrupts the gradient path tracing and CP localization algorithms.

Another question to address is how to handle the effects of the pseudopotential transformation and the missing core electrons. The valence pseudo-density is mostly coincident with the all-electron density in the bonding regions and interstitial, but smoother close to the nuclei, particularly for elements with few valence electrons or strongly cationic atoms. In these cases, maxima of the electron density may appear at positions other than the nuclei.

A transformation to the true all-electron density is possible, for instance, using the Projector Augmented Wave (PAW) datasets [80], but because the grid is not efficient at representing the steep density close to the core, it introduces an error in the integrated cell charge (and hence in the atomic charges). In CRITIC2 it is possible to augment the valence density by adding core densities from an internal database. The core densities must correspond to the correct number of core electrons, which are given by the user via the ZPSP keyword:

```
load file.cube
zpss C 4 H 1
```

The added core electrons dominate the behavior close to the nuclei and so, even though the direct addition of core densities is not theoretically justified, it is a very good approximation.

Because of the difficulties regarding gradient path tracing and CP localization, we recommend using alternatives to the traditional QTAIM methods when dealing with densities on a grid, such as the NCIPLOT technique for the topology or the Yu-Trinkle algorithm for the basin integration, both of which are also implemented in CRITIC2.

3.4. Computations involving fields

CRITIC2 is able to create new fields by performing arithmetic calculations on the existing ones. Some basic computation using FPLAPW fields is implemented (see the example in Section 5) but in the general case the newly created field is defined on a three-dimensional grid. The LOAD AS keyword is used for this task. For instance,

```
load as "$1+$2"
```

creates a grid containing the sum of the fields 1 and 2. CRITIC2 accepts the usual operators (+, −, *, /, modulo, exponential) and functions (sin, exp, erf, etc.). In addition, it is possible to use the higher-order derivatives of a field by inserting key letters after the dollar sign. For instance:

```
load as "0.01688686*$g1/$1^(4/3)"
```

loads a field corresponding to the reduced density gradient (Eq. (2)) by using the gradient (\$g) and the value (\$) of the electron density in field 1.

The LOAD AS keyword can also be used to perform grid integrations, particularly when combined with relational operators in the arithmetic expression. When a new grid is loaded, CRITIC2 calculates the cell integral, by summing over grid nodes. For example, the molecular volume is usually defined [81] as the volume of the region within the isodensity envelope defined by $\rho_v = 0.02 \text{ e/\AA}^3$. We can calculate the volume of a molecule in the urea crystal using the input:

```
crystal urea.cif
load as "($0 > 0.02*0.{529177}~{3}) / 2" 60 60 60
```

Field 0 is the promolecular density (in atomic units). The expression in parentheses returns a zero whenever the promolecular density is lower than ρ_v and one otherwise. The division by two accounts for the number of molecules in the cell. The three 60 are the grid size for the new field. The output is in atomic units:

```
[...]
Cell integral (grid SUM) ( 1) = 485.51169621
[...]
```

corresponding to a molecular volume of $71.9 \text{ \AA}^3$. Likewise, the charge per molecule in the interstitial is calculated using:

```
crystal urea.cif
load urea.cube
load as "($0 < 0.02*0.{529177}~{3})*$1/2"
```

where urea.cube is the electron density. The grid size for the new field is adopted from the density grid. The interstitial charge per molecule in the urea crystal is 0.123 electrons per molecule.

Any number of fields of any type can be combined in an arithmetic expression. Fig. 2 shows the deformation density (crystal minus promolecular density), the Laplacian of ρ , and the polarization density (crystal minus sum of *in-vacuo* molecular densities) for the urea crystal, together with the CRITIC2 inputs required to generate these plots. In all cases, the crystal structure and the density are read from a cube file calculated using Quantum ESPRESSO, and the density is core-augmented using the ZPSP keyword. The PLANE keyword is used in the all three inputs to plot a crystal plane encompassing the hydrogen-bond interactions, selecting the contour levels and mapping that look clearer in each case.

The LOAD AS keyword is the only essentially different command in the three inputs. In the first, the promolecular density (\$0) is subtracted from the crystal density (\$1). The plot shows the accumulation of electron density arising from intramolecular bond formation. In the second plot, the Laplacian is calculated using a Fourier transform (instead of spline interpolation), which yields much smoother plots. The shell structure around the heavy atoms is clearly visible, as well as the electron pairs around the oxygens. In the last input, the density of eight molecules surrounding the main cell has been calculated in the gas-phase using Gaussian. The eight cubes are read sequentially (fields 2 through 9) and then subtracted from the crystal density in PLANE (FIELD keyword). The electron density in the interstitial is mostly depleted (except in the hydrogen bond region) and accumulates near the nuclei in response to the polarizing effect of the molecules in the crystal. Fields created through LOAD AS behave exactly like an externally-loaded field and can be used in subsequent orders such as, for instance, basin integration (see Section 5).

4. Critical point search

The localization of critical points (CP) is a basic task in QTAIM and QCT. Different meanings have been attached to various types of CPs of the electron density and other fields. The first-order saddle points of the electron density (bond critical points, BCPs) are the definition of a chemical bond in QTAIM [6,7]. The value of the electron density at the BCP can be used to measure the relative strength of a bond. The value of the Laplacian at the BCP allows one to distinguish a covalent from a closed-shell interaction. The maxima of the ELF represent the regions where electrons are localized such as covalent bonds, lone pairs associated to electronegative atoms, and trapped electrons.

In order to access this information, the exact positions of the CPs of a field are necessary. In CRITIC2 the algorithm that performs the localization of CPs follows closely the one implemented in CRITIC. In the first step, the Wigner-Seitz (WS) cell is found, then partitioned into tetrahedra and reduced by symmetry to the irreducible

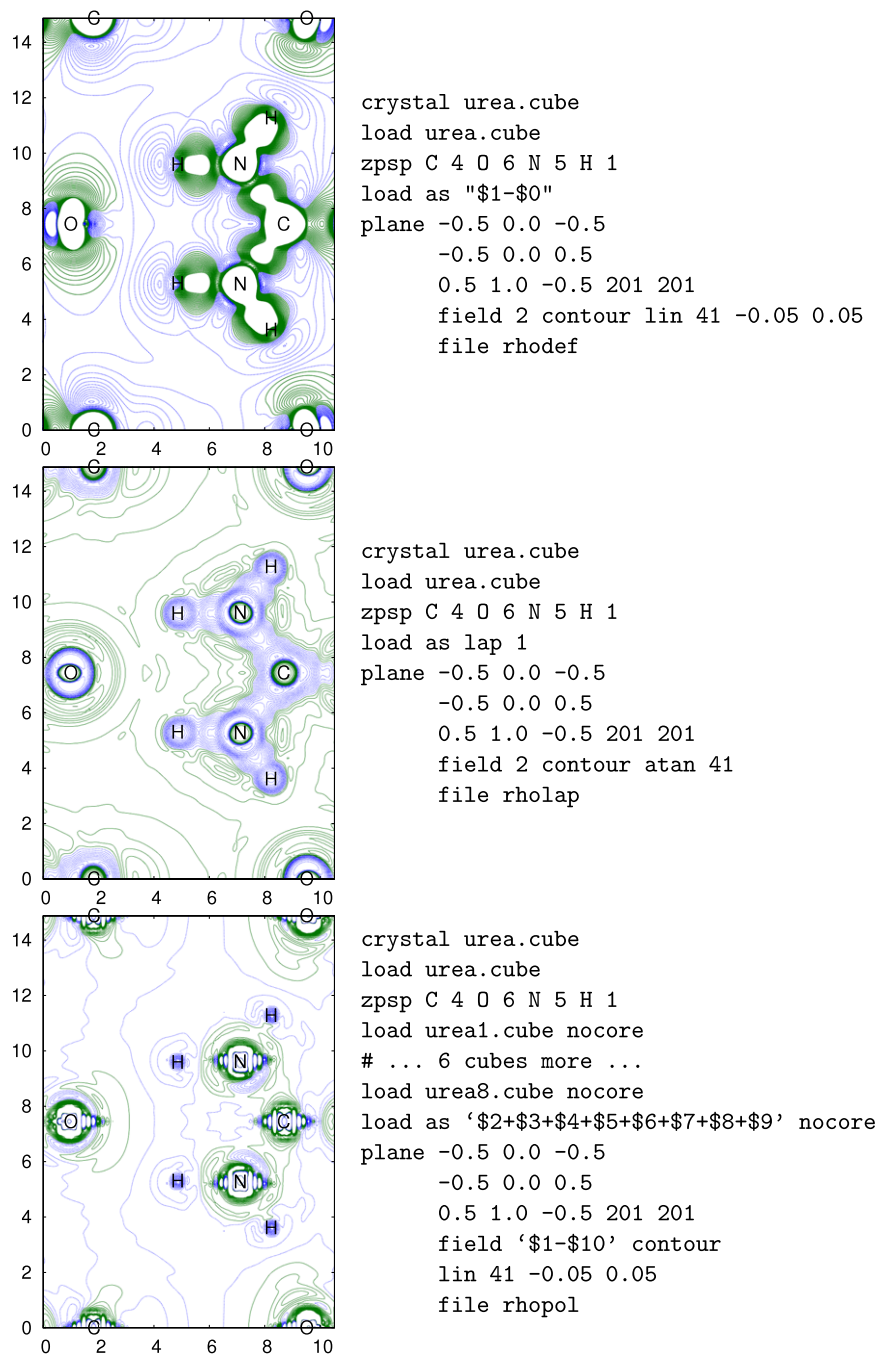


Fig. 2. Three representations of density-related quantities in a relevant plane of the urea crystal (origin $x_0 = (-1/2, 0, -1/2)$, x -axis $x_1 = (-1/2, 0, 1/2)$, y -axis $x_2 = (1/2, 1, -1/2)$). From top to bottom, we show the deformation density, the Laplacian and the polarization density. On the right, the inputs that generate the plots are shown. Positive contours are in green and negative contours are in blue. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Wigner–Seitz (IWS) cell. The symmetry-unique tetrahedra are recursively subdivided a user-defined number of times (typically one or two) and the vertices, edges, faces and interior of the final set of tetrahedra are used as seeds for the Newton–Raphson search. In addition, another seeding technique is used: the bond paths sometimes are difficult to find, especially if the field is noisy so seeds are placed at the midpoint between all atom pairs within 10 atomic units of each other.

The CP localization algorithm is invoked using the `AUTO` keyword. Fig. 3 shows the complete topology of the NPF_2 phosphazene (calculated using WIEN2k) which is generated using the input:

```
crystal npf2.struct
```

```
load npf2.clmsum npf2.struct
auto
```

Along with other information, `CRITIC2` prints a list of symmetry-unique CPs to the output, which in this case is:

```

* Critical point list, final report (non-equivalent cps)
topological class (n|b|r|c): 3( 16) 7( 48) 7( 52) 3( 20)
morse sum : 0

# n pg type position (cryst. coords.) mult name f |grad| lap
1 Cs (3,-3) nucleus 0.00000000 0.87100000 0.50000000 4 P 5+ 2.36625E+03 0.00000E+00 -3.00000E+15
2 Cs (3,-3) nucleus 0.00000000 0.08500000 0.70300000 4 N 3- 2.08844E+02 0.00000E+00 -3.00000E+15
3 C1 (3,-3) nucleus 0.12500000 0.70100000 0.56800000 8 F 1- 4.62646E+02 0.00000E+00 -3.00000E+15
4 C1 (3,-1) bond 0.93911819 0.58250528 0.14342024 8 b003 2.26342E-03 3.14074E-14 9.52803E-03
5 C1 (3,-1) bond 0.75583711 0.75824487 0.30716024 8 b005 2.26505E-03 2.45169E-16 1.12798E-02
6 C1 (3,-1) bond 0.61609051 0.00289927 0.31223744 8 b001 2.48521E-03 4.12676E-16 1.25246E-02
7 C1 (3,-1) bond 0.79904160 0.14290462 0.62559821 8 b006 3.05566E-03 6.68588E-16 1.35514E-02
8 C1 (3,-1) bond 0.05147393 0.80122344 0.52816532 8 b007 2.33087E-01 7.22020E-15 1.43081E+00

```

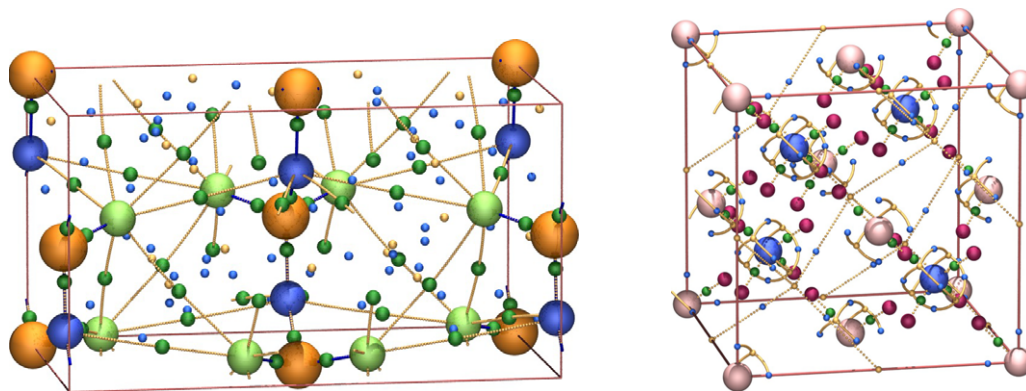


Fig. 3. Molecular graph of the electron density in the NPF₂ phosphazene (left) and of the ELF in cubic boron nitride (right). The color code for the atoms is: N (blue), P (orange), F (green), B (pink). The gradient paths are represented by golden lines. The critical points are shown as smaller balls: maxima (red), bonds (green), rings (blue) and cages (yellow). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

9 Cs (3, -1) bond	0.00000000	0.04421846	0.08444473	4 b004	2.42498E-01	8.06328E-16	7.73905E-01
10 Cs (3, -1) bond	0.50000000	0.39152404	0.37861790	4 b002	2.67810E-01	2.73809E-14	1.04436E+00
11 C1 (3, 1) ring	0.78218168	0.58147436	0.90196880	8 r005	9.24082E-04	7.59980E-16	4.64737E-03
12 C1 (3, 1) ring	0.31895209	0.13276906	0.11545254	8 r007	9.65206E-04	2.68843E-16	4.52237E-03
13 C1 (3, 1) ring	0.92153141	0.62008479	0.94283706	8 r003	1.18344E-03	4.53947E-16	5.13541E-03
14 C1 (3, 1) ring	0.69092797	0.66581481	0.40437348	8 r006	1.43513E-03	2.75833E-16	6.61734E-03
15 Cs (3, 1) ring	1.00000000	0.55988384	0.13277783	4 r001	1.99689E-03	1.87966E-13	8.68143E-03
16 C1 (3, 1) ring	0.91973922	0.43323956	0.72905958	8 r004	2.11253E-03	3.14981E-16	9.65156E-03
17 C1 (3, 1) ring	0.72244138	0.71358893	0.77176232	8 r002	2.21577E-03	1.08415E-13	1.06634E-02
18 C1 (3, 3) cage	0.67077198	0.85300689	0.47542756	8 c002	8.05798E-04	3.20440E-16	3.61286E-03
19 C1 (3, 3) cage	0.25521043	0.02869856	0.28980160	8 c003	8.50885E-04	1.35520E-14	4.01614E-03
20 Cs (3, 3) cage	1.00000000	0.58699078	0.90885911	4 c001	9.69295E-04	5.31279E-13	4.16436E-03

The number and type of CPs is given in the header (3 nuclei, 7 bonds, 8 rings and 3 cages) and the Morse sum is zero, indicating that the topology is most likely complete. A detailed list of properties at the symmetry-unique CPs follows, specifying their type and position, multiplicity, and the value of the density, gradient and Laplacian at the critical point. Bonds 8 through 10 present a density much higher than the rest and correspond to the direct covalent bonds. The remaining bonds belong to the interstitial space and have a small density, in the order of 10^{-3} atomic units.

The AUTO keyword can be used in combination with any scalar field, not necessarily the electron density. For instance, Fig. 3 (right) shows the topology of the ELF for the cubic BN crystal. The calculation was carried out with elk, and the ELF coefficients in Eq. (3) were written to an ELF.OUT file (the small modification to elk required to do this is included in the CRITIC2 distribution). The resulting file is passed to CRITIC2 using the input:

```
crystal GEOMETRY.OUT
load STATE.OUT GEOMETRY.OUT ELF.OUT
auto
```

In this case, the topology is not complete (the Morse sum is 32), probably because some of the core CPs are missing. Fig. 3 shows, however, that the relevant CPs of the ELF in the interstitial and around the atoms have been found. The shell structure of the ELF is apparent, with CPs surrounding both atoms and maxima of the ELF tetrahedrally coordinated around the anion.

5. Integration of attractor basins

The efficient integration of atomic basins is an outstanding challenge in QTAIM, particularly in crystals, where atomic basins usually present complex geometries [54]. The basin surfaces are determined by Eq. (1) and several integration algorithms, either with or without explicitly finding the IAS, have been proposed over the years [82–95,18–21]. Three methods are implemented in CRITIC2: bisection, qtree and the Yu–Trinkle (YT) algorithm.

The simplest approach is the bisection method [82]. A two-dimensional quadrature of the unit sphere is centered on every

symmetry-unique atom in the unit cell. Each quadrature node determines a ray originating at the position of the corresponding nucleus. The position of the IAS along that ray is determined by tracing upwards gradient paths, which terminate at a maximum. The point belongs in the basin only if the endpoint is the same as the atom being integrated. Bisection is the least efficient integration method in CRITIC2 but it is also robust provided the field being integrated is smooth. The bisection method is called using the INTEGRALS keyword. For instance, the input:

```
crystal quartz_o_DEN
load quartz_o_DEN
integrals gauleg 51 51
```

integrates the atomic charges and volumes in quartz (calculated using abinit) using a Gauss–Legendre spherical quadrature with $n_\theta = n_\phi = 50$ angular points. The atomic charges are 3.14 (Si), and -1.58 (O), with an integrated cell charge error of 0.025.

The QTREE algorithm [66] is based on a discretization of the smallest possible symmetry-unique region in the crystal, the irreducible Wigner–Seitz (IWS) region. The use of discretization method in QTAIM was first proposed by Malcolm and Popelier [96]. QTREE starts by finding the IWS, which is partitioned in tetrahedra. These tetrahedra are then recursively subdivided a given number of times (the subdivision level). The resulting tetrahedral grid nodes are assigned to the different attractors in the cell by tracing upwards gradient paths. Several tricks are used to speed up the assignment: (i) in-basin atomic integration spheres (β -spheres) are used to integrate the region closer to the nuclei, (ii) the nodes inside a tetrahedron whose vertices have been assigned to the same attractor are considered to be in the interior of the basin, and (iii) the gradient paths terminate when they penetrate a region that is known to be the interior of a basin. Because QTREE makes full use of crystal symmetry it is particularly well suited for small highly-symmetric crystals (fewer starting tetrahedra) and if the integrals of all basins are needed.

In CRITIC2, QTREE is invoked using the QTREE keyword. We illustrate it with a calculation of the magnetic moment of MnO calculated using WIEN2k:

```
crystal mno.struct
load wien mno.clmup mno1.struct
load wien mno.clmdn mno1.struct
load as clm add 1 2
reference 3
integrable "$1-$2"
qtree 5
```

The first line reads the crystal geometry from the struct file and the next two read the α - and β -spin densities. A LOAD AS

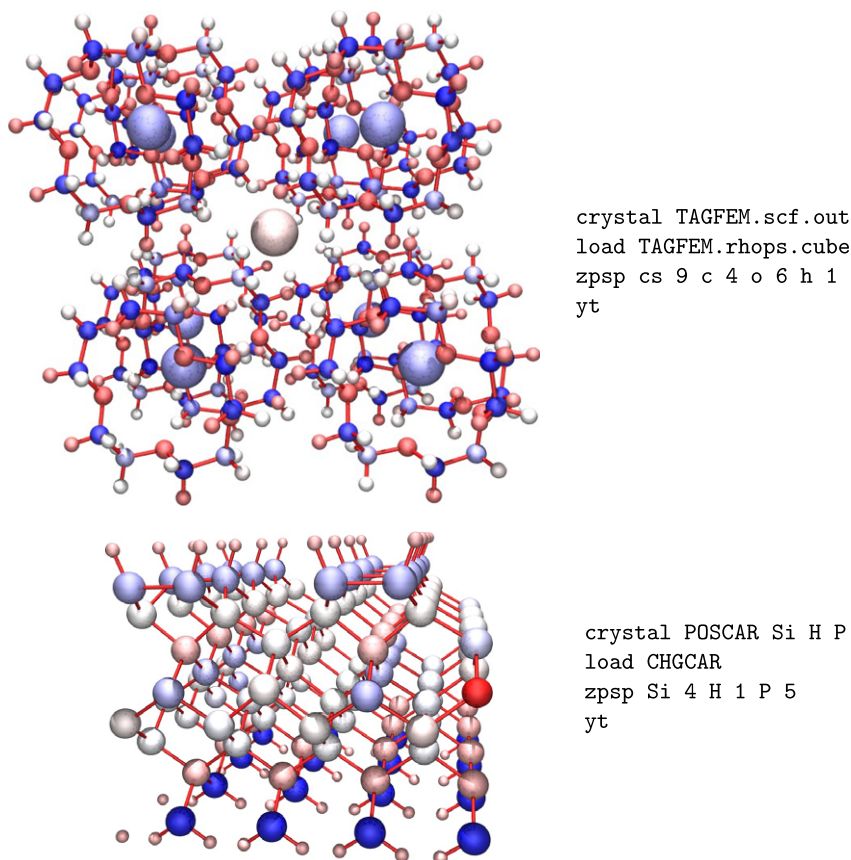


Fig. 4. Two example inputs showing the use of the YT keyword. Bader charges have been color-mapped on the atoms (blue: positive, red: negative). The first plot is an electride (bis(15-crown-5)-cesium) and the second is a phosphorus-doped hydrogen-terminated silicon surface. The color range limits are $[-m, m]$ with $m = 2.195$ for the electride and $m = 3.215$ for the silicon surface. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

keyword is used to build the total density in slot 3 (in FPLAPW format), which is then used as reference to obtain the integration basins (keyword REFERENCE). The properties to be integrated are defined using the INTEGRABLE keyword. By default, only volumes and charges (if the reference field is the electron density) are integrated. By using the \$1-\$2 expression, the spin density is added to the list of integrands, which will give the magnetic moments on integration. Finally, QTREE is called with 5 subdivision levels. The integrated spin density is 4.23 for two of the Mn atoms and -4.06 for the other two, corresponding to an antiferromagnetic state. The oxygens are non-magnetic with a spin density charge of 0.09 electrons.

The third integration method available in CRITIC2 is the Yu–Trinkle (YT) algorithm [21]. The YT algorithm belongs to a class of extremely efficient methods that use a grid representation of the electron density, first explored by Henkelman and co-workers [18–20]. Because a density grid is the outcome of a PSPW calculation, the combination of these two methods provide a very efficient route to the calculation of Bader charges in solids. In addition, the cost of these methods is linearly dependent on the size of the density grid and are only indirectly related to the number of atoms in the cell. As a consequence, grid-based methods are ideal (and practically the only option) for larger crystals.

In the YT algorithm, the grid points are sorted by decreasing electron density. A loop runs over all points in order, assigning them to the different atomic basins. The decision tree is:

- If a point has a density higher than all its neighbors then it is a local maximum. These usually correspond to nuclei but may appear at a certain distance from an atom because of the pseudopotential transformation of the density near the nuclei.

A distance criterion is used to assign maxima detected in this way to different atoms in the cell.

- If all the neighbors of a point belong to the same basin then the new point is considered to be in the interior of that basin.
- Otherwise, the volume represented by the point is considered to be straddled across an IAS.

The fraction of a point sitting on a IAS that belongs to a neighboring basin is calculated using an approximation to the trajectory flux out of that basin. This approximation depends on the density at the neighboring grid nodes as well as their geometric arrangement around the point being considered [21].

Fig. 4 shows two inputs using the YT integration method. The first is the bis(15-crown-5)-cesium electride. In this crystal an alkali metal (cesium in this case, represented by a large blue ball) is sandwiched between two crown ethers. The coordination with oxygen prompts the loose valence electron to migrate to the crystal void, depicted as a pink ball in the center of the plot. The cesium is almost cationic (charge 0.90) whereas the interstitial non-nuclear maximum contains 0.35 electrons. The rest of the charge is smeared over the crown ethers, where oxygen and carbon atoms are negatively and positively charged respectively.

The second example in Fig. 4 shows a hydrogen-terminated silicon surface which has been doped with a phosphorus atom. The P atom (red ball) stands out with a charge of -3.21 . The compensating positive charge, however, is surprisingly not localized on the atoms directly linked to the P but on the carbons at the bottom of the slab, colored in blue. This is probably an artifact of the hydrogen-capping. A slight alternation in charge of the silicon atoms can be seen throughout the bulk.

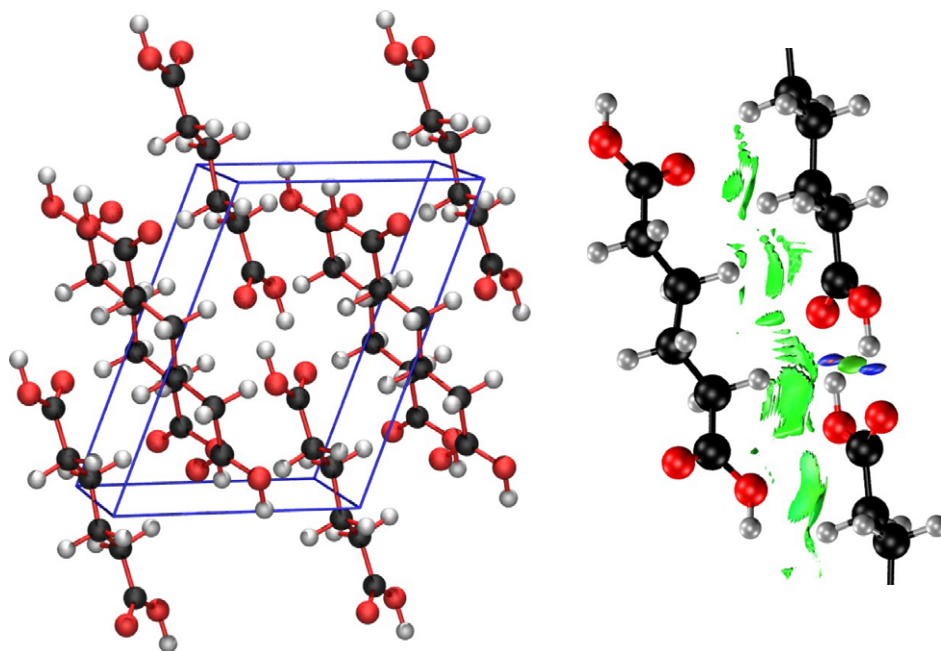


Fig. 5. Adipic acid crystal structure (left) and non-covalent interactions between three selected molecules at the crystal geometry (right). The NCI plot has been done using the $s = 0.5$ isosurface, and a density color scale from -0.05 to 0.05 . (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

In both examples, the input is very simple: the crystal structure is read (in the electrone from a Quantum ESPRESSO output, in the silicon surface from a POSCAR file) and then the density is loaded. The pseudopotential charge (ZPSP) is not strictly necessary because YT does not employ core-augmentation, but it facilitates reading the input since CRITIC2 can then calculate the charges using the valence atomic number. The YT method is invoked by using the YT keyword with no arguments. The algorithm is extremely efficient: it finds all the atomic charges in 14 s for the electrone (71 atoms in the unit cell) and in 84 s for the silicon surface (176 atoms) using a single Intel Core i7-2600 processor (3.4 GHz).

6. Non-covalent interactions index

The non-covalent interaction (NCI) index [38–40] is a simple and computationally cheap way of examining molecular close contacts and non-covalent interactions. There are two basic objects in NCI: the electron density (ρ) and the reduced density gradient (RDG, s), the latter given by Eq. (2). The key to identifying interaction regions is that they present low s (hence they are close to a critical point of the density) and low ρ (there is no covalent bond) at the same time [38]. By representing the isosurfaces of low s in the regions with small value of ρ it is possible to obtain a map of density overlap, which is known to correspond to intermolecular Pauli repulsion [40].

The region to be explored for non-covalent interactions is chosen at the beginning of the run (by default, the entire unit cell). A three-dimensional grid of density and RDG is calculated spanning that region. Four files are written in the NCI task: the density cube, the RDG cube, the data file (containing a list of s vs. ρ on all the grid points) and a vmd script for the visualization of the calculated isosurfaces. The simplest way to execute this task is by invoking the NCIPLOT environment:

```
crystal rho.cube
load rho.cube
nciplot
endnciplot
```

If the system is too large to calculate the self-consistent electron density, it is possible to run NCIPLOT with the promolecular density by simply omitting the LOAD keyword.

Molecular crystals are usually tightly packed [81] and, as a consequence, using the entire cell in NCIPLOT produces an overabundance of isosurfaces that obscure the interpretation of the plot. The region can be restricted by giving its coordinates manually, but it is usually more convenient to define molecular fragments and examine only the NCI regions which lie between those molecules. We illustrate this feature using the adipic acid crystal in Fig. 5. Adipic acid forms molecular chains via a double hydrogen bond between facing carboxylic acid groups. The inter-chain interaction is much weaker, involving the hydrophobic body of the molecule. A direct plot of all the NCI regions in the cell would be very busy, so we first write an xyz file containing the molecules in the main cell using:

```
crystal rho.cube
write adipic_acid.xyz molmotif
```

Then, we open the adipic_acid.xyz file with a molecular visualizer and extract the coordinates of three molecules as shown in the right part of Fig. 5. These three molecules are representative of the main interactions in the crystal. The cartesian coordinates of the fragments (in angstrom) are then passed back to CRITIC2 using the FRAGMENT keyword. For a pair of molecules:

```
crystal rho.cube
load rho.cube nocore
nciplot
fragment
0.38067      3.73058      8.15806
# ...
-1.19405     6.18790     5.49396
endfragment
fragment
2.86622      3.04398      5.90002
# ...
4.22789      1.32636     10.18065
endfragment
endnciplot
```

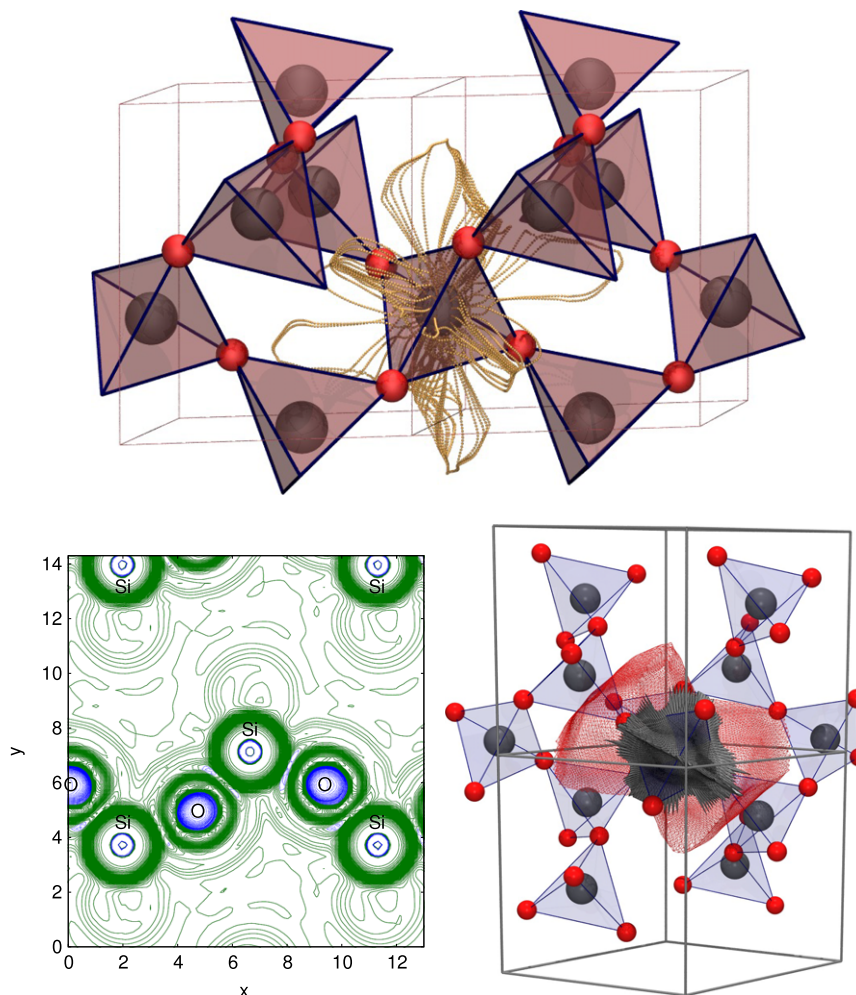


Fig. 6. Several plots for the α -quartz (SiO_2) crystal are shown. Top: the crystal structure with a sampling of gradient paths originating at a Si atom. Bottom left: the Laplacian of the electron density (atomic units); positive Laplacian contours are green and negative are blue. Bottom right: the basins of the three symmetry-unique atoms; two oxygens (red) and one silicon (black). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Using this input, the region where the density and RDG are mapped is automatically reduced to encompass only the chosen fragments, and the NCI regions are plotted only if they lie between the two molecules. The plot in Fig. 5 can be generated by having an NCIPLOT block for each of the three pairs in the trimer. The hydrogen bonds appear clearly as blue (i.e. high density) domains whereas the weak inter-chain interaction appears as an extended low density (green) region.

7. Graphical representations

CRITIC2 implements the graphical representation capabilities of the previous version [53]. The list of possible representation styles includes evaluations of an arbitrary field at points, lines, planes, and three-dimensional regions, gradient path tracing, contour plotting on a plane and determination of attractor basins and primary bundles. We illustrate several possible representations in Fig. 6 for the α -quartz phase of SiO_2 . The input is:

```
crystal quartz_o_DEN
load quartz_o_DEN
fluxprint
  ncp 7 10 10
endfluxprint
grdvec
plane -0.2 0.5 0.3 1.2 0.5 0.3 -0.2 0.5 1.7
```

```
contour lap atan
endgrdvec
basinplot method cube 5 output basin
```

The FLUXPRINT keyword traces and represents gradient paths originating at arbitrary points in the unit cell. In the example, the atom (nuclear CP) number 7 (Si) is used as the origin to trace downwards gradient paths using a spherical mesh with $n_\theta = n_\phi = 10$. The resulting plot is shown at the top of Fig. 6: the Si basin clearly exceeds the coordination polyhedron and squeezes between the neighboring oxygens forming very sharp wings that end at the interstitial cage. Using GRDVEC it is possible to make gradient path and contour representations of an arbitrary field on a plane. In the example, the Laplacian (LAP keyword) of the density is mapped on a plane (the origin, x-axis and y-axis are given in groups of 3 crystallographic coordinates) using an arctangent transformation (Fig. 6, bottom left). The Si–O interaction is mostly ionic and therefore closed shell (the Laplacian at the bond critical point is small and positive) and the map clearly shows the shell structure around each atom.

The atomic basins can be represented using the BASINPLOT keyword. The procedure is similar to bisection integration: a polyhedron is placed around a nucleus (in this case a cube) and then subdivided a number of times (5 in the input). The limit of the IAS is located by gradient path tracing and bisection. Fig. 6 (bottom right) shows the three non-equivalent basins in quartz. The red basins are the oxygens, showing the characteristic cushion shape of

the anions. The basin of the Si cation is curved inwards and presents narrow wings that stretch to the density cages.

8. Parallelization, installation and user's guide

Parts of the CRITIC2 code have been parallelized for shared-memory architectures (OpenMP). The parallelization applies to:

- In AUTO, the CP search algorithm is parallelized over seeds.
- Bisection, both for basin integration and plotting, is parallelized over rays.
- When a new grid is created in LOAD and CUBE keywords, the process is parallelized over grid nodes. This applies to the generation of the promolecular density grid in NCIPLLOT as well.
- QTREE is parallelized over IWS tetrahedra.

The installation procedure has been ported to autotools. The distribution contains a full-fledged user's manual and syntax guide (in the doc/ subdirectory) and a set of tests and examples (in examples/). Some auxiliary programs to manipulate the files generated by CRITIC2 (basin and off files, for instance) are also provided in the tools/ subdirectory.

9. Conclusion

In this article, we have presented CRITIC2, the new version of the CRITIC program for the analysis of chemical interactions in real space. CRITIC2 reads crystal structures and scalar fields (e.g. the electron density), in a variety of formats, including those natively generated by WIEN2k, elk, PI, abinit, Quantum ESPRESSO, VASP, Gaussian and practically any program that can generate a field on a grid. Arithmetic operations can be carried out to generate new fields from existing ones and to perform cell integrations.

CRITIC2 applies common techniques to the loaded scalar field, irrespective of their origin, traditional in QTAIM and QCT analysis: localization of critical points (CP), gradient path tracing and basin integration. The CP localization algorithm is based on a recursive subdivision of the symmetry-unique portion of the Wigner–Seitz cell. Gradient path tracing allows the calculation of interatomic surfaces, that can be either plotted or integrated for atomic properties. Three basin integration algorithms are available: bisection, QTREE and Yu–Trinkle. The latter is grid-based and is extremely efficient, enabling the computation of atomic properties for very large systems in minutes on a single processor.

The NCIplot technique, an alternative to QTAIM topology, is also implemented in CRITIC2. It maps the regions of molecular (non-covalent) contact in a crystal by plotting the isosurfaces of the reduced density gradient. Because no CP localization is involved, the procedure is less complex (and more robust) and is ideally suited for densities defined on a grid.

Finally, the code has been parallelized for shared-memory architecture, it is fully documented and examples are provided with the distribution. CRITIC2 is distributed under free (GNU/GPL) license.

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